भारतीय सूचना प्रौद्योगिकी संस्थान भागलपुर
Indian Institute of Information Technology Bhagalpur
Department of Electronics & Communication Engineering

End Semester, Autumn 2023
B. Tech. 3rd Year, Semester V
Subject: EC304 IoT & Embedded System

Duration: 180 Minutes **Maximum Marks: 50**

Instructions:
Draw appropriate figure wherever necessary.
Assume necessary data wherever required.
Figures in the right margin indicate marks.
Assume crystal frequency to be 11.0592 MHz.

1.

a) Write an ARM assembly program to compute the factorial of a positive number (>1) stored in R0. Save the result in R1. **[3]**

b) With the help of a suitable block diagram explain the embedded system. **[4]**

c) Differentiate between von Neumann and Harvard architectures used in Embedded Systems. **[3]**

2.

a) Discuss applications of IoT for the realization of smart cities. **[3]**

b) Discuss the link layer IoT protocols. **[3]**

c) Show the interfacing of 16 × 2 LCD with 8051 microcontroller and write an embedded C program to display your name on the interfaced LCD. **[4]**

3.

a) Write a suitable program to transfer the letters of first field of your name serially as per the below specified baud rate continuously.
If your Roll No. is even transfer serially at a baud of 9600.
If your Roll No. is odd transfer serially at a baud of 4800. **[4]**

b) Write a suitable program to generate a square wave signal of frequency 1 kHz. Use timer, T1 in mode 2. **[3]**

c) Explain DB-9 connector. **[3]**

4.

a) Explain pre-emptive shortest job first (SJF) scheduling algorithm. **[3]**

b) With the help of suitable example explain programming of timer interrupt. **[4]**

c) Explain PCA timer. **[3]**

5.

a) Explain the significance of real-time system in the embedded system design. **[3]**

OR

b) Explain how M2M is different from IoT. **[3]**

c) Discuss the features of Python and explain how it motivated to use it for developing IoT systems. **[4]**

d) What do you mean by Internet of Things (IoT)? Explain the characteristics of IoT. **[3]**